Mathematical Modeling Lecture Notes
Chapter 9 – Decision Analysis

I. Introductory problem – Silicon Dynamics
   a. Hand out copies of problem.
   b. Should the company sell the rights to the chip, or should it market and produce the computer itself?
      i. There are some different ways we could think about this.
      ii. Some of these depend on the likelihood of the computer catching on.
   c. Tree representation of problem
      i. For Silicon Dynamics

      ii. Two types of nodes
         1. Decision nodes
            - Represented by squares
            - Indicate that a decision needs to be made
         2. Chance nodes
            - Represented by circles
            - Indicate that a random event occurs

II. Decision-making criteria
   a. Maximum-likelihood criterion
      i. This criterion says to base the decision on what is most likely to happen.
      ii. Applying this criterion to the Silicon Dynamics problem
         1. If the probability of the computer catching on is larger than 0.5, then we should market and produce the computer.
            - In this case, we'll earn $0 if the computer does not catch on and $54 million if it does.
         2. Otherwise, we should sell the rights to the computer.
      iii. Pros and cons of this criterion
         1. Pro: You're basing your decision on what is most likely to happen.
         2. Con: For the action you choose, there may be a less likely state of nature with a really bad payoff.
         3. Con: For an action you don't choose, there may be a huge possible payoff with a small probability.
         4. Con: If the most likely situation is not much more likely than some others, does it really make sense to focus on that?
   b. Maximin criterion
      i. This criterion can be viewed in terms of game theory.
      ii. It says that for each possible decision alternative, find the minimum payoff over all possible states of nature. Then find the maximum payoff over all the decision alternatives, and make that decision.
      iii. Applying this criterion to the Silicon Dynamics problem
         1. If we sell the rights, we make $15 million automatically.

2. If we market and produce the computer...
   - We'll make \$0 if the computer doesn't catch on and \$54 million if it does.
   - The state of nature with the minimum payoff is that the computer doesn't catch on, in which case we make \$0.
3. The maximum minimum outcome for the two choices is the \$15 million we get for selling the rights.
4. So that's what we should do according to the maximin criterion.

iv. Pros and cons of this criterion
1. Pro: This criterion gives the best guarantee of a payoff.
2. Pro/con: It's a strongly risk-averse strategy.

c. Bayes's decision rule
i. This criterion says to make the decision with the best expected payoff, assuming prior probabilities for each state of nature.
ii. Applying this criterion to the Silicon Dynamics problem
1. Selling the rights earns an automatic \$15 million.
2. If p is the probability that the computer catches on, then the expected value of marketing and producing the computer is $54p$, in millions of dollars.
3. Thus if $54p > 15$, or $p > 5/18 \approx 0.28$, then we should market and produce the computer.
4. If $p < 5/18$, then we should sell the rights.

iii. Pros and cons of this criterion
1. Pro: It incorporates all available information.
2. Pro: If you were faced with making the same decision over and over again, this criterion would result in the best decisions in the long run (because of the law of large numbers).
3. Con: If the prior probabilities are wildly inaccurate, then this may result in a poor decision.

iv. This is the most widely used criterion and the one we will stick with for the rest of our discussion of decision analysis.
v. Simple sensitivity analysis
1. Graph of p versus the expected amount of money for each decision

2. What and how good is our estimate of p?
   - If our estimate of p is much larger or much smaller than the crossover point, then we can be satisfied that our decision is a good one.
   - If our estimate of p is close to the crossover point, then we need to be fairly sure that p is accurate in order to be confident in our decision.

d. See text, Section 9.4, for some other decision-making criteria.

III. Decision making with experimentation
a. Idea
i. Sometimes you can spend additional money to gain more information on the likelihood of the states of nature.

ii. Is it worth it to do this?

b. Back to the Silicon Dynamics example
   i. Suppose it would be possible to spend $1 million to do market research that would predict which of the two levels of demand is most likely to occur. Such research is correct two-thirds of the time.
   ii. If we do the market research then all we know is what the market research indicates, and then we have to make a decision.
   iii. This leads to the following decision tree:

   iv. To finish the decision tree we need the following probabilities:
      1. P(Computer Catches On | Favorable Market Research)
      2. P(Computer Doesn't Catch On | Favorable Market Research)
      3. P(Computer Catches On | Unfavorable Market Research)
      4. P(Computer Doesn't Catch On | Unfavorable Market Research)
      5. P(Favorable Market Research)
      6. P(Unfavorable Market Research)
      7. The first four are called the *posterior probabilities*.
   v. Determining the probabilities
      1. To answer these questions it helps to draw the probability tree (not the decision tree), which describes the states of nature.

      2. For simplicity's sake, let's take the prior probability of the computer catching on to be 0.4.

- We could vary these later to do some sensitivity analysis.

3. P(Favorable Market Research) is $1/5 + 4/15 = 7/15$.
4. P(Unfavorable Market Research) is $8/15$.
5. P(Computer Catches On | Favorable Market Research) can be answered:
   - Out of all the cases in which the market research is favorable, which ones are those in which the computer catches on?
   - Probabilistically, this is

$$\frac{\frac{4}{15}}{\frac{1}{5} + \frac{4}{15}} = \frac{\frac{4}{15}}{\frac{7}{15}} = \frac{4}{7}.$$

   - This is an application of *Bayes's Theorem*, which we you should have seen in probability theory: If events $B_1, B_2, \ldots, B_m$ are mutually exclusive, $\cup_{i=1}^{m} B_i = S$, $P(A) > 0$, and $P(B_i) > 0$ for each i, then, for any k,

$$P(B_k|A) = \frac{P(B_k \cap A)}{P(A)} = \frac{P(B_k)P(A|B_k)}{\sum_{i=1}^{m} P(B_i)P(A|B_i)}.$$

6. Thus P(Computer Doesn't Catch On | Favorable Market Research) $= 3/7$.
7. Similarly, P(Computer Catches On | Unfavorable Market Research) is

$$\frac{\frac{2}{15}}{\frac{2}{15} + \frac{2}{5}} = \frac{1}{4}.$$

8. P(Computer Doesn't Catch On| Unfavorable Market Research) $= 3/4$.

vi. Now we have the following decision tree.

vii. The optimal policy using Bayes's decision rule is not to perform the market research and attempt to market and produce the computer. The expected payoff in this case is \$21.6 million.

viii. How much is the information from the market research really worth?

1. If our estimates of the probabilities are correct, and the market research didn't cost us anything, then the expected payoff for doing the market research would be $22.4 million.
2. Since the expected payoff when not doing the market research is $21.6 million, the value to us of the market research is $800,000.
3. Thus it would be in our interest to spend the money on the market research if it costs less than $800,000 and not worth it if the market research costs more than $800,000.

IV. Utility theory
  a. Introduction
    i. Decision analysis with Bayes's decision rule uses the expected monetary payoff as the appropriate measure.
      1. If you make the same decision over and over, this is usually a good measure because your losses and gains will average to the expected payoff over time.
      2. However, if you often only have to make the decision once, and a bad decision might be catastrophic, then this measure might not be a good one.
    ii. There is another way to measure decisions, though: the *utility* of money, rather than its expected value.
    iii. Utility can be thought of as the value of money relative to the decision-maker.
    iv. A practical thought experiment: Would you rather have $1 million guaranteed or a 50% chance at $2 million?
      - The expected monetary payoff criterion would treat these as equivalent, but for some people and some situations they may not be.
  b. Utility functions
    i. Three basic types
      1. Declining marginal utility
        - More money is worth relatively less and less (negative second derivative)
        - Most people feel this way about money, for the most part — if they have enough to buy certain things the extra isn't as important.
        - Having this type of utility function is considered *risk averse*.
        - Graphically:

      2. Increasing marginal utility
        - More money is worth relatively more and more (positive second derivative)
        - Example: Someone who is trying to raise capital for a particular reason.
        - Having this type of utility function is considered *risk seeking*.
        - Graphically:

      3. Constant marginal utility
        - More money is worth, relatively speaking, the same amount as little money (zero second derivative)

- This is the utility function inherent in the traditional use of Bayes' decision rule.
- Having this type of utility function is considered *risk neutral*.
- Graphically:

ii. How utilities work
   1. The fundamental idea is that a decision maker is indifferent between two actions with the same expected utility.
   2. Example
      - Imagine a person with the following utilities:
        - $U(\$100,000) = 1$.
        - $U(\$150,000) = 2$.
        - $U(\$180,000) = 3$.
        - $U(\$200,000) = 4$.
      - This means that \$200,000 is worth four times as much to this person as \$100,000 is.
      - Suppose there were a 25% chance of gaining \$200,000.
        - The expected utility of this option would be $0.25(4) = 1$.
        - This means that having \$100,000 for certain is worth the same to the person as having a 25% chance of obtaining \$200,000.
   3. Essentially, the monetary value is replaced by its utility measure when calculating expected values.

iii. Applying utilities to the Silicon Dynamics problem
   1. Suppose the utility function for Silicon Dynamics is $U(x) = \sqrt{x}$ (a classic function with a negative second derivative).
   2. How would that change the policy or decision for Silicon Dynamics?
   3. The decision tree now looks like this:

4. The optimal decision is "Don't do the market research. Sell the rights."

Homework: Chapter 9

- Section 9.1 #6, 7
- Section 9.2 #2, 6
- Section 9.3 #3, 5, 7
- Section 9.4 #6